

Electromagnetic Listening & Ambient Transduction: Capturing Inaudible Parasitic Frequencies in Built Environments

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ABSTRACT

Modern urban architectures are inundated with invisible, inaudible electromagnetic radiation emitted by power lines, switching power supplies, cellular arrays, and computer microprocessors. In this monograph, we present custom wideband inductive receiver coils and balanced preamplifiers designed to sonify these parasitic electromagnetic fields. We outline methods for mapping urban electromagnetic soundscapes, isolating rhythmic clock pulses from industrial machinery, and incorporating stray electronic fields into live electroacoustic performance.

KEY THEOREMS & BOUNDARY EQUATIONS

Formulation [1.1]

Principle of Inductive Audio Transduction: Time-varying magnetic fields induce proportional micro-voltages across high-turn copper coils, translating electromagnetic radiation directly into audio signals without radio demodulation.

Formulation [2.1]

Parasitic Harmonics Hierarchy: Switching-mode power supplies produce predictable odd-harmonic series whose fundamental frequency fluctuates under electrical grid loading.

Formulation [3.1]

Spatial Electromagnetic Cartography: Moving directional loop receivers through architectural spaces maps invisible functional boundaries corresponding to concealed wiring and server installations.

1. The Inaudible City

Beneath the audible soundscape of traffic and human conversation lies an intense, unseen electromagnetic infrastructure. Every transformer hums, every LED driver oscillates at tens of kilohertz, and every microprocessor radiates microscopic clock pulses. By rendering these signals audible, sound artists gain access to a hidden sensory dimension of our built world.

2. Receiver Coil Engineering

Standard guitar pickups and tele-coils lack the wideband frequency response needed for clean electromagnetic listening. We engineered dual-axis orthogonal coils wound with 5,000 turns of 44 AWG enameled copper wire around high-permeability mu-metal cores. Coupled to an ultra-low-noise differential JFET preamplifier, the system achieves a flat response from 5 Hz to 45 kHz.

3. Rhythmic and Timbral Structures

Electromagnetic soundscapes exhibit unexpected musical structures: rhythmic ostinatos from server memory refresh cycles, melodic chirps from motor speed controllers, and atmospheric textures from long-wave atmospheric radio emissions. We demonstrate real-time granular synthesis techniques that transform these raw electromagnetic hums into expressive musical pads and percussive hits.

4. Archival and Field Recordings

A catalog of electromagnetic recordings gathered across power stations, subway networks, and computer server centers is archived at ZIAA, providing a unique historical record of the electronic pulse of twenty-first-century infrastructure.

REFERENCES & PRIMARY CITATIONS

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